purchases. Near real-time analytical results may be one reason to consider using streams.

The second reason to consider streams in the data world is to use event sourcing, known from the software development world. Event sourcing stores the changes associated with an entity (e.g., a transaction) rather than the final state of the entity. Having all the changes from the very beginning provides tremendous opportunities for seeing an entity at any point in time, such as building any representation based on them, creating analyses that were not considered when the system was created, and time travel. The stream in this form might be the basis for creating other data products.

It is worth noting that a stream can also be used to implement a data product —to transport the data needed to build the data product. However, it's then considered an input interface.

VISUALIZATIONS

Our last example of a port is a visualization: various kinds of charts and dashboards representing data. They can be a good resource for nontechnical users, especially if they are combined with a filtering and data selection mechanism. This port provides visual representations of data in a relatively easy way. Such automatic visualizations can be supported by one of the platform capabilities.

Table 5.5 summarizes port applications for various situations.

Table 5.5 Characteristics of the data product ports

Type of data product port	Access type	Amount of data
Database-like storage	SQL-like interface Programmatically	Big amount of data
Files	Scripting	Small to big